FYS5419/9419 March 18

FYS5419/9419 March 18

Mathematics of Discrete Fourier transforms & quantum Fourier transforms (QFT)

DFT (Discrete Fourier transf)

$$|x\rangle = \vec{x} = \begin{bmatrix} x_0 \\ x_1 \\ \vdots \\ x_{n-1} \end{bmatrix}$$

$$|y\rangle = U|x\rangle$$

x_js are complex numbers

$$|y\rangle = \begin{bmatrix} y_0 \\ y_1 \\ \vdots \\ y_{n-1} \end{bmatrix}$$

$$y_k = \frac{1}{\sqrt{n}} \sum_{j=0}^{n-1} x_j \, e^{2\pi i (jk)/n}$$

$$|y\rangle = \frac{1}{\sqrt{n}} \begin{bmatrix} \displaystyle\sum_{j=0}^{n-1} x_j \, e^{2\pi i (j \cdot \underset{k=0}{0})/n} \\ \displaystyle\sum_{j=0}^{n-1} x_j \, e^{2\pi i (j \cdot 1)/n} \\ \vdots \\ \displaystyle\sum_{j=0}^{n-1} x_j \, e^{2\pi i (j(n-1))/n} \end{bmatrix}$$

$$x^T = [1, 2] \qquad ([x_0\ x_1])$$

$$y^T = [y_0\ y_1]$$

$$y_0\ (k=0) = \underbrace{\frac{1}{\sqrt{2}}}_{x_0} + \underbrace{\frac{2}{\sqrt{2}}}_{x_1} = 3/\sqrt{2}$$

$$y_1\ (k=1) = \frac{1}{\sqrt{2}} + \frac{2}{\sqrt{2}} \underbrace{e^{i\pi}}_{-1}$$

$$|y\rangle = U|x\rangle \wedge U = \frac{1}{\sqrt{2}}\begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}$$

$$U \in \mathbb{C}^{N\times N} \sim O(N^2)$$

$$UU^{\dagger} = U^{\dagger}U = \mathbb{1}$$

FFT $\sim O(N \log N)$

Digression: polynomial expansion

$$f(x_i) \simeq P_{n-1}(x_i)$$

$$= \sum_{j=0}^{n-1} a_j x_i^j = a_0 x_i^0 + a_1 x_i^1 + \dots a_{n-1} x_i^{n-1}$$

$$|f\rangle = \begin{bmatrix} P(x_0) \\ P(x_1) \\ \vdots \\ P(x_{n-1}) \end{bmatrix} = V|a\rangle$$

$$V = \begin{bmatrix} 1 & x_0^1 & x_0^2 & \cdots & x_0^{n-1} \\ 1 & x_1^1 & x_1^2 & \cdots & x_1^{n-1} \\ \vdots & & \ddots & & \\ 1 & & & & x_{n-1}^{n-1} \end{bmatrix}$$

$$|a\rangle = \begin{bmatrix} a_0 \\ a_1 \\ \vdots \\ a_{n-1} \end{bmatrix}$$

We get something similar with DFT

$$\omega = e^{2\pi i/N}$$

$$x_0^1 = e^{2\pi i(0\cdot 1)/N} = 1$$

$$x_0^2 = e^{2\pi i(2\cdot 0)/N} = 1$$

$$x_1^1 = e^{2\pi i/N} = \omega$$

$$x_1^2 = e^{2\pi i(2\cdot 1)/N} = \omega^2$$

$$U = \frac{1}{\sqrt{N}} \begin{bmatrix} 1 & 1 & - & - & 1 \\ 1 & w & w^2 & w^3 & \cdots & w^{N-1} \\ 1 & w^2 & w^4 & - & \cdots & w^{2(N-1)} \\ \vdots \\ 1 & w^{N-1} & - & \cdots & w^{(N-1)^2} \end{bmatrix}$$

$$y_k = \frac{1}{\sqrt{N}} \sum_{j=0}^{N-1} x_j \, e^{2\pi i (j \cdot k)/N}$$

$$|y\rangle = U|x\rangle$$

QFT

$$\langle x_i | x_j \rangle = \delta_{ij}$$

$$\langle y_i | y_j \rangle = \langle x_i | \underbrace{U^\dagger U}_{\mathbb{1}} | x_j \rangle$$

$$= \delta_{ij}$$

N is the dim of $|x\rangle$ and $|y\rangle$

$$U \in \mathbb{C}^{N \times N}$$

with the qubits

$N = 2^n$ $\qquad n = \#$ qubits

Example 1 $\qquad n = 1 \; ; \; N = 2$

$|0\rangle \wedge |1\rangle$

$$|y\rangle = U|x\rangle$$

$$= \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix} \begin{bmatrix} |0\rangle \\ |1\rangle \end{bmatrix}$$

$$|\tilde{0}\rangle = \frac{1}{\sqrt{2}} \sum_{k=0}^{1} e^{2\pi i\, 0\, k/2} |k\rangle$$

$$= \frac{1}{\sqrt{2}} \left[e^{2\pi i\, 0 \cdot 0/2} |0\rangle + |1\rangle \right]$$

$$= \frac{1}{\sqrt{2}} \left[|0\rangle + |1\rangle \right]$$

$$|\tilde{1}\rangle = \frac{1}{\sqrt{2}} \left[|0\rangle + e^{\pi i} |1\rangle \right]$$

$$= \frac{1}{\sqrt{2}} \left[|0\rangle - |1\rangle \right]$$

$$|x\rangle \longrightarrow \boxed{H} \longrightarrow |y\rangle$$

Example 2 : $N = 2^2 = 4$

$n = 2$

$$|y\rangle = \left\{ \begin{array}{cccc} |0\rangle \otimes |0\rangle, & |0\rangle \otimes |1\rangle, & |1\rangle \otimes |0\rangle, & |1\rangle \otimes |1\rangle \\ |00\rangle & |01\rangle & |10\rangle & |11\rangle \end{array} \right\}$$

$$|x\rangle \qquad \rightarrow |0\rangle \qquad |1\rangle \qquad |2\rangle \qquad |3\rangle$$

$$|y\rangle = QFT\,|x\rangle$$

$$|y\rangle = |y_0 y_1\rangle = |01\rangle$$

$$(|y_1 y_0\rangle)$$

integer binary rep

$$y = 2^{n-1} y_0 + 2^{n-2} y_1 + \ldots$$

$$+ 2^0 y_{n-1}$$

$n = 2$

$$y = 2^1 y_0 + 2^0 y_1$$

$$y_0 = 0 \quad y_1 = \underline{1} \Rightarrow y = \underline{1}$$

$|01\rangle \rightarrow |1\rangle$

$|11\rangle \rightarrow |3\rangle$

$$2^1 y_0 + 2^0 y_1 = 3$$

$$y_0 = 1, \quad y_1 = 1$$

Example 3 $\quad n = 3 \quad N = 8$

$|000\rangle, \; |001\rangle, \; |010\rangle \; \ldots \; |111\rangle$

$|000\rangle = |0\rangle \otimes |0\rangle \otimes |0\rangle = |0\rangle$

$|001\rangle \rightarrow |1\rangle$

$|111\rangle \rightarrow |7\rangle$

$$y = 2^2 \underbrace{y_0}_{=0} + 2^1 \underbrace{y_1}_{=0} + 2^0 \underbrace{y_2}_{=1} = 1$$

$N = 4 \; ; \; n = 2$

$$|j\rangle = \frac{1}{2} \sum_{j=0}^{3\,(4-1)} e^{2\pi i\, jk/N} \, |k\rangle$$

$$U = \begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & i & -1 & -i \\ 1 & -i & 1 & -i \\ 1 & -i & -1 & i \end{bmatrix}$$

$$|00\rangle \longrightarrow \frac{1}{2}\left(|0\rangle + |1\rangle + |2\rangle + |3\rangle\right)$$

$$|0\rangle = \frac{1}{2}\left(|00\rangle + |01\rangle + |10\rangle + |11\rangle\right)$$

$$= \frac{1}{2}\begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix}$$

$$= \frac{1}{2}\begin{bmatrix} e^{i2\pi j/2^0} \end{bmatrix}$$

$$\otimes \begin{bmatrix} e^{2\pi i j/2^1} \end{bmatrix}$$

$$J = J_0 2^{n-1} + J_1 2^{n-2} + \cdots$$

$$J_{n-2} 2^{1} + J_{n-1} 2^{0}$$

$$\left(\begin{array}{l} |00\rangle \to |0\rangle \\ |01\rangle \to |1\rangle \\ |10\rangle \to |2\rangle \\ |11\rangle \to |3\rangle \end{array} \right.$$

$$K = k_0 2^{n-1} + k_1 2^{n-2} + \cdots k_{n-1} 2^{0}$$

$$|J_0 J_1 \cdots J_{n-1}\rangle = |J\rangle$$

$$|k_0 k_1 \cdots k_{n-1}\rangle = |K\rangle$$

$$|j\rangle = \frac{1}{2^{n/2}} \sum_{k_0=\{0,1\}} \sum_{k_1=\{0,1\}} \cdots \sum_{k_{n-1}=\{0,1\}}$$

$$\times\ e^{i 2\pi j \sum_{m=0}^{n-1} k_m 2^{-m}}$$

$$\times\ |k_0 k_1 \ldots k_{n-1}\rangle$$

$$= \frac{1}{2^{n/2}} \sum_{k_0=\{0,1\}} \sum_{k_1=\{0,1\}} \cdots \sum_{k_{n-1}=\{0,1\}} \bigotimes_{m=0}^{n-1} e^{i 2\pi j k_m 2^{-m}} |k_m\rangle$$

$$= \frac{1}{2^{n/2}} \bigotimes_{m=0}^{n-1} \left\{ \sum_{k_m=\{0,1\}} e^{i2\pi j k_m / 2^m} |k_m\rangle \right\}$$

$n = 2$

$$|j_0 j_1\rangle = \frac{1}{2} \left(|0\rangle + e^{i2\pi j_0/2} |1\rangle \right)$$

$$\otimes \left(|0\rangle + e^{i2\pi j_1/2^2} |1\rangle \right)$$

$$|00\rangle = \frac{1}{2} \left(|00\rangle + |01\rangle + |10\rangle + |11\rangle \right)$$

Binary floats instead of binary integers

$$\left(y = y_0 2^{n-1} + y_1 2^{n-2} + \dots y_{n-1} 2^0 \right)$$

$$|y\rangle = |y_0 y_1 \dots y_{n-1}\rangle$$

$n = 2$

$$0.y_1 = y_1/2$$

$$0.y_0 y_1 = y_0/2 + y_1/4$$

$$0.y_k y_{k+1} \dots y_{n-1} = y_k 2^{-1} + y_{k+1} 2^{-2} + \dots +$$

$$y_{n-1} \; 2^{-(n-k)}$$

$$n = 2$$

$$|y\rangle = |y_0 y_1\rangle$$

$$|y\rangle \longrightarrow \frac{1}{2}\left[|0\rangle + e^{2\pi i\, 0.y_1} |1\rangle \right]$$

$$\otimes \left[|0\rangle + e^{2\pi i\, 0.y_0 y_1} \right]$$

$$y_0 = y_1 = 0 \qquad U|x\rangle$$

For two qubits:

on the first qubit we have a Hadamard gate, and the second one is affected by a rotation gate